Fire Hazard

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All About Fire

Fire is a chemical reaction with three essential elements which are well-represented as a triangle. These elements are:

- Fuel
- Air
- Heat

Fire is a useful element and force of nature because it produces light and heat. However, fire can reduce to ashes an entire house and building, and even people in a few hours. It can also destroy an entire plantation due to harvest and hectares of forest. As defined by the National Fire Protection Association (n.d.), fire is "a rapid oxidation process, which is a chemical reaction resulting in the evolution of light and heat in varying intensities". Also, it is a corrosion process that occurs very quickly. The light, heat, and sound are released swiftly which causes the temperatures to rise.

The three components necessary to create fire include fuel (something that will burn), heat (enough to make the fuel burn), and air (oxygen). This is also called the fire triangle. Eliminating fire requires removal of one or more of its components" (National Fire Protection Association, n.d.).

Common Causes of Fire and Tips for Prevention

The South Australian Country Fire Service (n.d.) identified ten common causes of fire and provided tips for preventing fire. These are presented in the table below.

Causes of Fire	***Tips to Prevent Fire***
Kitchen Stoves	• Never leave the stove unattended. • Check if electric cords, curtains, tea towels, and oven cloths are at a safe distance from the stovetop. • Be careful of long flowing sleeves contacting gas flames.
Electric Blankets	• Do not sleep with electric blankets on or leave the house without switching them off. • Never leave weighty objects on the bed when the electric blanket is on. • Have your blanket checked by an authorized repairer if you suspect overheating. • Always follow the manufacturer's instructions for care and storage. • Inspect each blanket for wear and tear at the beginning of the cooler months.
Faulty Wiring	• Always hire a qualified electrician. • Double adaptors and power boards can overload power points. • Install safety switches and correct fuses.
Smoking in Bed	• Smoking in bed can be fatal - tiny embers can smolder unnoticed and burst into flame much later.
Lighting	• Check light fittings for heat build-up. • Discard lampshades that are close to light globes and lamp bases that can be knocked over easily. • Ensure recessed downlights are properly insulated from wood paneling or ceiling timbers.
Flammable Liquids	• Store all flammable liquids such as petrol, kerosene, and methylated spirits away from heat. • Always check the label before use and storage. • Use extreme care when pouring.

Clothes Dryers	• Always clean lint filters after each load. • Avoid drying bras in your dryer as the underwire can get caught and start a fire.
Candles	• Never leave burning candles unattended. Do not sleep with a burning candle. • Make sure curtains and other flammable items are well away from burning candles.
Home Heating	• Make sure all appliances are professionally installed. • Check that walls and floors are insulated from heat sources. • Be careful where you place portable appliances. • Never leave an open fire alight when you leave the house or go to bed. • Place a mesh guard in front of open fires. • Have your chimney and flue cleaned annually. • Never leave children unattended near fires and heaters. • Clothing should not be dried close to heaters or fires.
Children	• Warn all children about playing with fire. • Keep all matches, lighters, and candles out of reach of small children. • Teach young children to bring matches or lighters they find to an adult immediately. • Teach older children that matches are a tool to be used only in the presence of adults. • Brief your babysitter on your fire plan. Make sure they know all exits and emergency telephone numbers. • Make sure that the babysitter understands fire survival techniques.

What to Do When There is a Fire

Before

1. Observe fire prevention guidelines in your community or school premises.
2. Fire produces a thick and heavy smoke which will prevent you from seeing your surroundings.
3. Thereby at the preparatory period or before a fire, it would help if you prepare an exit plan.
4. Know the post-evacuation plans of your location.
5. Ensure a fire extinguisher.
6. Look for fire exits and memorize floor plans.

During

1. Do not panic.
2. Do not run.
3. Do not use elevators.
4. Find the emergency alarm button if you are inside an establishment or school and press it to sound.
5. Protect yourself from the fire.
6. Call the fire department.
7. Cover your nose and drop. Crawl to the nearest fire exit accessible from your location.
8. If you tried to extinguish the fire and it cannot be handled, put the extinguisher down and leave the fire.
9. Feel the doorknobs and cracks with the back of your hand before opening any door.
10. Do not forget to close the doors and windows that you used to pass through as you escape from the fire. This will delay the spread of fire.
11. Get out as quickly as you can and stay away from smoke and falling objects.

After

1. Call the fire department if they are not yet alerted.
2. Once you are out of the burning premises or building, never attempt to go back to save things or lives. Let the authorized personnel do it for you.
3. Once you are already outside, go to a safe place, check yourself for injuries, and stay alert.
4. Stay away from the building that has been damaged by fire unless the authorities already allowed people to enter the area.

Other Fire-Fighting Safety Techniques and Procedures

Fire Inside a Building

- Get out immediately when there is a fire inside your building.
- Fire produces a thick and heavy smoke which will prevent you from seeing your surroundings. At the preparatory period or before a fire, it would help if you prepare an exit plan.
- Have a fire plan and practice it. This plan should include fire drills. A fire drill can help you practice leaving your building quickly and safely.
- Do not forget to talk about different ways you can get out.

Fire Inside a Room with the Door Closed

- Do not open the door. If the door is hot, there is smoke coming through the cracks around or under the door, or the doorknob is hot, this means that the fire is close.
- If you are trapped in a room during a fire, use duct tape, wet towels, or clothing to seal off any cracks or vents around the door. Call an emergency number and let them know where you are. You could even put a light-colored cloth (like a white t-shirt) outside the window to let firefighters know where you are.
- If the doorknob feels cool and smoke is not coming from the room, open the door slowly and carefully. Then quickly exit the building.
- If a fire starts in your kitchen because of oil or grease, toss baking soda or salt on it or put a lid over the flame if the fire is in a pot or pan. Do not use water to extinguish a burning pan.

How to Extinguish a Fire Using a Fire Extinguisher

1. Pull the pin
2. Aim the nozzle
3. Squeeze the lever
4. Sweep

Observance of the Fire Prevention Week

The annual observance of Fire Prevention Month during March is based on Presidential Proclamation 115-A issued on November 17, 1966. It is meant to provide education and awareness during the summer months to lessen the incidence of fire in the country. How can we inspire and remind Filipinos to take precautions to prevent fires at homes, in offices, and in different communities under the theme "*Kaligtasan sa Sunog: Alamin, Gawin, at Isabuhay Natin*".

If you want to help in case of a fire incident, you can provide direct help in one of the ways identified below. It is, however, important to take note that you can only do this if you have the ability and knowledge in firefighting.

- Provide background logistical support
- Inform the local fire station
- Help in crowd control and in keeping the curious onlookers at bay
- Help in directing fire engines and rescue vehicles (particularly in the internal lanes/by lanes, etc.)
- Clear the way for the firefighting crew
- Do not put yourself and others at risk
- Do not question the established chain of command. There are multiple ways of fighting a fire. This is not the time to argue and convince each other on the best method.

Reference:

Lanada, M.I.B., Melegrito, M.L.F., & Mendoza, D.J. (2016). *Building resilient communities: Disaster readiness and risk reduction.* Phoenix Publishing House